Code for creating a dashboard including:

Revenue Analysis
Profit Analysis
Shipment Status Tracking
Inventory Monitoring
Sales Trend Analysis
Product Performance
Regional Revenue Analysis
KPI Cards
Logistics & Delivery Metrics
The script includes the same dashboard concepts such as:
✔ Data Cleaning
✔ Missing Value Handling
✔ Duplicate Removal
✔ KPI Calculations
✔ Revenue Charts
✔ Shipment Distribution
✔ Monthly Sales Trends
✔ Top Product Analysis
✔ Inventory Turnover
✔ Delivery Performance
The KPI formulas in the code are aligned with the business metrics shown in your dashboard, including:
Total Revenue
Total Profit
Profit Margin
Total Orders
Delivery Rate
Inventory Turnover
Product Performance
Shipment Analysis

Source code

```
# PYTHON CODE FOR SUPPLY CHAIN DASHBOARD

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt


# LOAD DATASET
```

```
df = pd.read_csv(
    r"C:\Users\Naveen\Downloads\Supply_Chain_Dashboard\supply_chain_dashboard.csv"
)


# DISPLAY DATASET

print("FIRST 5 ROWS")
print(df.head())

print("\nDATASET INFORMATION")
print(df.info())


# DATA CLEANING

# Remove duplicate rows
df = df.drop_duplicates()

# Fill missing values
df = df.fillna(0)

# Remove extra spaces from column names
df.columns = df.columns.str.strip()


# DATE CONVERSION

if 'Order Date' in df.columns:
    df['Order Date'] = pd.to_datetime(df['Order Date'])

# KPI CALCULATIONS

# Total Revenue
if 'Revenue' in df.columns:
    total_revenue = df['Revenue'].sum()
else:
    total_revenue = 0

# Total Cost
if 'Cost' in df.columns:
    total_cost = df['Cost'].sum()
else:
    total_cost = 0
```

```
# Total Profit
total_profit = total_revenue - total_cost

# Profit Margin
if total_revenue != 0:
    profit_margin = (
        total_profit / total_revenue
    ) * 100
else:
    profit_margin = 0

# Total Orders
if 'Order ID' in df.columns:
    total_orders = df['Order ID'].nunique()
else:
    total_orders = 0

# Total Customers
if 'Customer ID' in df.columns:
    total_customers = df['Customer ID'].nunique()
else:
    total_customers = 0

# Total Products
if 'Product ID' in df.columns:
    total_products = df['Product ID'].nunique()
else:
    total_products = 0

# Total Quantity
if 'Quantity' in df.columns:
    total_quantity = df['Quantity'].sum()
else:
    total_quantity = 0

# Average Order Value
if total_orders != 0:
    average_order_value = total_revenue / total_orders
else:
    average_order_value = 0

# Shipment Analysis
if 'Shipment Status' in df.columns:
```

```
    delayed_shipments = df[
        df['Shipment Status'] == 'Delayed'
    ].shape[0]

    delivered_shipments = df[
        df['Shipment Status'] == 'Delivered'
    ].shape[0]

    in_transit_shipments = df[
        df['Shipment Status'] == 'In Transit'
    ].shape[0]

else:

    delayed_shipments = 0
    delivered_shipments = 0
    in_transit_shipments = 0

# On-Time Delivery Rate
if (
    delivered_shipments +
    delayed_shipments
) != 0:

    on_time_delivery_rate = (
        delivered_shipments /
        (
            delivered_shipments +
            delayed_shipments
        )
    ) * 100

else:

    on_time_delivery_rate = 0

# Inventory Turnover
if 'Current Stock Level' in df.columns:

    inventory_turnover = (
        total_revenue /
        df['Current Stock Level'].sum()
    )
```

```
else:
    inventory_turnover = 0

# DISPLAY KPI RESULTS
print("\n========== KPI RESULTS ==========")
print("Total Revenue :", total_revenue)
print("Total Cost :", total_cost)
print("Total Profit :", total_profit)
print("Profit Margin % :",
      round(profit_margin, 2))
print("Total Orders :", total_orders)
print("Total Customers :", total_customers)
print("Total Products :", total_products)
print("Total Quantity Sold :", total_quantity)
print("Average Order Value :",
      round(average_order_value, 2))
print("Delivered Shipments :",
      delivered_shipments)
print("Delayed Shipments :",
      delayed_shipments)
print("In Transit Shipments :",
      in_transit_shipments)
print("On-Time Delivery Rate % :",
      round(on_time_delivery_rate, 2))
print("Inventory Turnover :",
      round(inventory_turnover, 2))
```

```
print("=================================")

# CHART 1 : REVENUE BY REGION


if 'Region' in df.columns and 'Revenue' in df.columns:

    revenue_region = df.groupby(
        'Region'
    )['Revenue'].sum()

    plt.figure(figsize=(10,5))

    revenue_region.plot(kind='bar')

    plt.title("Revenue by Region")

    plt.xlabel("Region")

    plt.ylabel("Revenue")

    plt.xticks(rotation=45)

    plt.show()

# CHART 2 : SHIPMENT STATUS DISTRIBUTION


if 'Shipment Status' in df.columns:

    shipment_status = df[
        'Shipment Status'
    ].value_counts()

    plt.figure(figsize=(7,7))

    shipment_status.plot(
        kind='pie',
        autopct='%1.1f%%'
    )

    plt.title("Shipment Status Distribution")
```

```
    plt.ylabel("")

    plt.show()

# CHART 3 : MONTHLY SALES TREND


if 'Order Date' in df.columns and 'Revenue' in df.columns:

    monthly_sales = df.groupby(
        df['Order Date'].dt.month
    )['Revenue'].sum()

    plt.figure(figsize=(10,5))

    monthly_sales.plot(
        kind='line',
        marker='o'
    )

    plt.title("Monthly Sales Trend")

    plt.xlabel("Month")

    plt.ylabel("Revenue")

    plt.grid(True)

    plt.show()

# CHART 4 : TOP PRODUCTS BY REVENUE

if 'Product Name' in df.columns and 'Revenue' in df.columns:

    top_products = df.groupby(
        'Product Name'
    )['Revenue'].sum().sort_values(
        ascending=False
    ).head(10)

    plt.figure(figsize=(12,5))

    top_products.plot(kind='bar')
```

```
    plt.title("Top 10 Products by Revenue")

    plt.xlabel("Product Name")

    plt.ylabel("Revenue")

    plt.xticks(rotation=45)

    plt.show()

# CHART 5 : SALES BY SHIPMENT STATUS

if 'Shipment Status' in df.columns and 'Revenue' in df.columns:

    shipment_sales = df.groupby(
        'Shipment Status'
    )['Revenue'].sum()

    plt.figure(figsize=(8,5))

    shipment_sales.plot(kind='bar')

    plt.title("Sales by Shipment Status")

    plt.xlabel("Shipment Status")

    plt.ylabel("Revenue")

    plt.xticks(rotation=0)

    plt.show()

# CHART 6 : INVENTORY ANALYSIS

if 'Product Name' in df.columns and 'Current Stock Level' in df.columns:

    inventory_data = df.groupby(
        'Product Name'
    )['Current Stock Level'].sum().sort_values(
        ascending=False
    ).head(10)

    plt.figure(figsize=(12,5))
```

```
    inventory_data.plot(kind='bar')

    plt.title("Top 10 Products by Inventory Level")

    plt.xlabel("Product Name")

    plt.ylabel("Current Stock Level")

    plt.xticks(rotation=45)

    plt.show()

# SAVE CLEANED DATASET


df.to_csv(
    r"C:\Users\Naveen\Downloads\cleaned_supply_chain_dashboard.csv",
    index=False
)

print("\nDashboard Analysis Completed Successfully!")
```

Python Script visual in Power BI

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

# Load Dataset
df = pd.read_csv(
    r"C:\Users\Naveen\Downloads\Supply_Chain_Dashboard\supply_chain_dashboard.csv"
)

# Remove duplicate rows
df = df.drop_duplicates()

# Fill missing values
df = df.fillna(0)

# Remove extra spaces from column names
df.columns = df.columns.str.strip()
```

```
# Total Revenue KPI
total_revenue = df['Revenue'].sum()

# Total Profit KPI
total_profit = df['Revenue'].sum() - df['Cost'].sum()

print("Total Revenue :", total_revenue)
print("Total Profit :", total_profit)
```

Example

1. Revenue by Region Chart

```
import pandas as pd
import matplotlib.pyplot as plt

# Group Revenue by Region
revenue_region = df.groupby(
    'Region'
)['Revenue'].sum()

# Create Bar Chart
plt.figure(figsize=(8,5))

revenue_region.plot(kind='bar')

plt.title('Revenue by Region')

plt.xlabel('Region')

plt.ylabel('Revenue')

plt.xticks(rotation=45)

plt.show()
```

2. Shipment Status Distribution Chart

```
import pandas as pd
import matplotlib.pyplot as plt

# Count Shipment Status
shipment_status = df[
```

```
    'Shipment Status'
].value_counts()

# Create Pie Chart
plt.figure(figsize=(6,6))

shipment_status.plot(
    kind='pie',
    autopct='%1.1f%%'
)

plt.title('Shipment Status Distribution')

plt.ylabel('')

plt.show()
```

3. Monthly Sales Trend Chart

```
import pandas as pd
import matplotlib.pyplot as plt

# Convert Order Date
df['Order Date'] = pd.to_datetime(
    df['Order Date']
)

# Monthly Revenue Analysis
monthly_sales = df.groupby(
    df['Order Date'].dt.month
)['Revenue'].sum()

# Create Line Chart
plt.figure(figsize=(8,5))

monthly_sales.plot(
    kind='line',
    marker='o'
)

plt.title('Monthly Sales Trend')

plt.xlabel('Month')
```

```
plt.ylabel('Revenue')

plt.grid(True)

plt.show()
```

<u>Output</u>